

FORENSIC DOC ANALYZER v1.0.0

TECHNICAL DOCUMENT & USER MANUAL

Author & Signature : Dam-FOR3K (with the help of Antigravity AI) • Version 1.0.0

Platform : 100% Local, Standalone and Air-Gapped Solution (No Internet)

■ **PURPOSE OF THE TECHNICAL MANUAL :** This document is the technical user guide for **Forensic Doc Analyzer v1.0.0**. It details the inspection engines used for each file format (PDF, DOCX, XLSX, EML, MSG, OLE), the meaning of detected anomalies, and evidence traceability (Source Location).

1. Philosophy & Risk Scoring System

The application assigns a **Forensic Risk / Suspicion Index** to each analyzed document, scaled from **0 to 100**.

Range	Risk Level	Meaning & Recommended Actions
0 — 24 pts	LOW / NORMAL	Standard document with no critical anomalies. Healthy structure.
25 — 59 pts	MODERATE	Presence of atypical elements (e.g., orphan RSID, visual redaction, un-obfuscated macro). Requires a review of the production chain.
60 — 100 pts	CRITICAL	Major alerts (Executable OLE payload, Follina/CVE exploit, remote UNC template, sub-second timestamping). Risk of attack or malicious forgery.

Warning regarding Category Caps (CATEGORY_MAX_CAPS): To avoid score saturation (100/100) on legitimate documents with multiple minor repetitions, the tool applies a maximum cap of assignable points per category (e.g., Orphan RSID capped at 20 pts max, Third-party Author at 10 pts max).

2. Analysis Engines & Format Interpretation

2.1. Microsoft Word Documents (.docx, .docm, .doc)

A. Author vs. Modifier Metadata:

- *dc:creator* (Original Creator) vs *cp:lastModifiedBy* (Last Modifier).
- **Interpretation:** A discrepancy between the creator and the last modifier is common in corporate environments. However, if the creator is anonymous or generated by an automated tool (e.g., 'User', 'Admin'), this is an indicator of anonymization.
- ■ **Evidence Source :** docProps/core.xml -> <dc:creator> & <cp:lastModifiedBy>

B. RSID (Revision Save IDs) Editing Session Mapping:

- **What is an RSID?** An RSID is an 8-character hexadecimal identifier generated by Word during each save operation.
- **Orphan RSID (undeclared):** Presence of an RSID in the text (word/document.xml) that is **NOT** listed in the official register word/settings.xml.
- **Forensic interpretation:** This is irrefutable proof that this text segment was **copied and pasted from an external Word document** or injected by a third-party script.

- ■ **Evidence Source** : word/document.xml -> Attribute w:rsidR (absent from word/settings.xml)

C. Recent Uniform ZIP Timestamps (Word 2021/365 in 2026):

- **Interpretation:** In the historical OOXML standard, internal XML files are frozen at the MS-DOS ZIP epoch (1980-01-01). However, during a 'Save As...' in Word 2021/365 on Windows 11, Word applies the current save date to all XML files. **If all XMLs share the same recent uniform date (e.g., 2026-08-21), the document is structurally legitimate.** The alert only triggers in case of abnormal heterogeneity.

D. Remote Template (attachedTemplate UNC / Web):

- **Risk:** Template Injection attack. The document loads a .dotm template hosted on a remote server (e.g., \\rogue_server\malware.dotm or https://...).
- ■ **Evidence Source** : word/_rels/settings.xml.rels -> Target='...'

E. Fake Redaction & Camouflaged Text:

- **Fake Redaction:** A black geometric shape (<v:rect fillcolor='black'>) was drawn over the text to mask it visually, but the original text remains fully readable in the underlying XML code.
- **Camouflaged Text:** Text configured in white color (<w:color w:val='FFFFFF'>) or in micro-font (< 2pt).
- ■ **Evidence Source:** word/document.xml

F. Language Detection & Suspicious Transitions:

- Analyzes the declared language (w:lang) for every run of text to identify suspicious language transitions (which may indicate localized tampering or localized paragraph forgery).

G. Document Protection & Restrictions:

- Searches for active editing restrictions (w:documentProtection), enforcement modes (read-only, forms), and password hashing algorithms used to secure the document.

H. Hidden Text & SDT Controls:

- Locates w:vanish tags (text hidden from the GUI but present in binaire) and Structured Content Controls (w:sdt elements) which may carry orphaned metadata.

I. Internal Comments History:

- Fully extracts comments from the word/comments.xml file (including author names, creation dates, and raw text content), which are often forgotten during cleanup.

2.2. PDF Documents (.pdf)

A. Structure & Suspicious Keywords (PDFiD):

- Search for active executable objects: /JavaScript, /JS, /AA (Auto-Action), /OpenAction, /Launch, /EmbeddedFiles.
- ■ **Evidence Source** : PDFiD Python Engine -> PDF object analysis

B. Incremental Revision Markers (%%EOF):

- **Interpretation:** A standard PDF file ends with a single %%EOF marker. If it contains multiple %%EOF markers, the document has undergone incremental revisions. The tool extracts each previous version to verify if pages or signatures were retrospectively masked or altered.

C. PDF Collusion Tool (2-File Binary Comparison):

- Allows comparing a suspicious PDF against its presumed original to extract the exact difference matrix (modified texts, deleted or added objects).

D. Cryptographic Signature Verification (PyHanko):

- Inspects the validity of the X.509 trust chain of digital certificates and validates the RFC 3161 timestamp.

2.3. Microsoft Excel Documents (.xlsx, .xlsm, .xls)

A. Chronological Text Sequence (sharedStrings.xml):

- In the XLSX format, all text strings entered by users are stored in a centralized dictionary (xl/sharedStrings.xml) in the chronological order of their entry.
- **Detecting Orphaned (Deleted) Texts:** Even if a string value is deleted or updated in a worksheet, Excel does not purge it from sharedStrings.xml immediately. The application cross-references sharedStrings.xml indices against actively referenced sheet cells and isolates 'orphaned strings'.
- **Forensic Interpretation:** Reconstructs historical entries, deleted rows, or overwritten data, showing the exact chronological sequence of data entry.
- **Evidence Source:** xl/sharedStrings.xml vs xl/worksheets/sheet*.xml (orphan index mapping)

B. Formula & Calculation Cache Analysis:

- Analyzes and extracts active formulas (<f> tags) and cross-references them with cached pre-calculated values (<v> tags). This helps detect manual edits of cached cell contents without corresponding formula updates.

C. Calculation Chain Validation (calcChain.xml):

- Cross-validates entries in the calculation chain (xl/calcChain.xml) against active formulas. Discrepancies can reveal orphaned or hidden formulas in the workbook.

D. Hidden Sheets (veryHidden) & Absolute Paths:

- Detects hidden worksheets set to the veryHidden state (which cannot be unhidden via standard Excel GUI). Extracts the original local absolute path of the workbook (x15ac:absPath), revealing the author's folder structure.

2.4. Image Retouching Analysis (Error Level Analysis - ELA)

- **ELA Principle:** Error Level Analysis re-saves the image at a 95% JPEG compression level and calculates the error difference pixel by pixel.
- **Result interpretation:**
 - *Uniform / dark zones:* The entire image shares the same compression rate (Original untouched image).
 - *Very bright / locally highly contrasted zones:* Indicates a recent insertion, montage, or collage originating from another image with a different compression rate.
- **Automatic RGB JPEG conversion:** Seamless support for non-JPEG images (PNG, BMP, TIFF, WebP).

2.5. Emails & Outlook Messages (.eml, .msg)

A. MSG OLE Compound File Structure:

- Outlook .msg files are OLE binary containers. The tool natively extracts metadata streams (*Sender, To, Subject, Date, Message-ID, Message Class*).

B. Header Authentication (SPF, DKIM, DMARC):

- Analysis of the Authentication-Results line and raw network headers (*Internet Headers*) to validate the authenticity of relay servers.

C. Outlook FILETIME Decoding (Thread-Index):

- Extracts the raw timestamp of the initial message in the conversation from the Base64 Thread-Index header to validate the actual sending server time.

D. Text / HTML Divergence (Phishing):

- Calculates the length discrepancy between the Plain-Text and HTML versions of the message. A difference > 150 characters indicates hidden text in HTML used to evade anti-spam filters.

E. MSG Attachment Extraction & Quarantine:

- Secure extraction of all embedded attachments within the MSG file, including SHA-256 fingerprint calculation and a quarantine download button.

2.6. Stylometry & ZIP Steganography

- **Lexical richness (Type-Token Ratio - TTR):** Measures vocabulary variety. A TTR < 25% indicates an automated or highly repetitive text.
- **Average sentence length:** Calculated by splitting on punctuation AND line breaks (`\n` / `\r`).
- **ZIP Steganography:** Detects hidden files within the ZIP container that are not declared in the [Content_Types].xml table.

2.7. Credits & Forensic Libraries (33 Monitored Dependencies)

- **Didier Stevens:** PDF Tools (pdfid, pdf-parser) — blog.didierstevens.com
- **Philippe Lagadec (decalage2):** oletools (olevba, oleid, oletimes, oleobj, msodde, rtfobj, oledir, olebrowse) — decalage.info
- **Phil Harvey:** ExifTool — exiftool.org
- **Matthias Valcke:** PyHanko — github.com/MatthiasValcke/pyHanko
- **Poppler & QPDF:** Stream extraction, fonts & structural checks (14 binary executables).
- **Dam-FOR3K (with Antigravity AI):** Design & Interface

3. Input Methods & System Date Preservation (OS)

The application offers two methods to analyze a document, each directly impacting the disk system metadata (*FileModifyDate*, *FileAccessDate*, *FileCreateDate*) extracted by ExifTool:

A. Browser Upload Mode (Drag & Drop):

- **Mechanism:** For security reasons, web browsers strip local disk dates during transfer. The application subsequently saves a temporary file on the analyst's machine.
- **Forensic Impact:** The system dates (*FileModifyDate*) will display the exact time of the analysis (e.g., today at 16:25). *Note: The true internal dates of the document (PDF CreateDate, docProps) remain strictly intact.*

B. Direct Local Path Mode (Recommended):

- **Mechanism:** The analyst enters the absolute path of the file (e.g., `C:\Cases\Contract.pdf`). The application reads the file directly from the hard drive WITHOUT creating a temporary file.
- **Forensic Impact:** ExifTool extracts the **true original system dates from the disk**. The Forensic Timeline thus accurately includes the *FileModifyDate* timestamp of the original local file, guaranteeing perfect OS chronology integrity.

4. Evidence Traceability (Source Location) & Mapping

Each indicator flagged by the tool includes its ■ **Source Location** field. Here is a reference summary:

Anomaly Category	Evidence Location (Source Location)	Role in the Report
Web Converter / Apple Pages	docProps/app.xml -> <Application> PDF Metadata -> /Producer	Proves the original host software.
Orphan RSID	word/document.xml -> Attribute w:rsidR	Proves a copy-paste from an external source.
Remote Template Attachment	word/_rels/settings.xml.rels -> Target	Proves a UNC/Web Template Injection attempt.
Sub-Second Timestomping	docProps/core.xml -> <dcterms:modified>	Proves manual date tampering (.000000).
Fake Redaction	word/document.xml -> <v:rect fillcolor='black'>	Proves that visually masked text is present in clear text.

■■ **Methodological Forensic Disclaimer:**

IMPORTANT: This software is an experimental exploration and triage assistant. In digital forensics, no automated output should be accepted as definitive evidence without manual corroboration. Forensic examiners must independently verify all extracted events, timestamps, and anomalies by directly inspecting the primary source containers using independent reference tools.